Supplementary Materials for

A brain-computer interface that evokes tactile sensations improves robotic arm control

Sharlene N Flesher PhD[1,2,3], John E Downey PhD[1,2,3,4], Jeffrey M Weiss MS[1,5], Christopher L Hughes MS[1,2,3], Angelica J Herrera BS[1,2,3], Elizabeth C Tyler-Kabara MD[6], Michael L Boninger MD[1,2,5,7,8], Jennifer L Collinger PhD[1,2,3,5,8]*†, Robert A Gaunt PhD[1,2,3,5]*†.

Correspondence to: rag53@pitt.edu and collinger@pitt.edu

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Movies S1 to S4
Data File (Bidirectional_BCI_ScoreandTimes_Data.xlsx)

Materials and Methods

Participant

The male participant was 28 years old at the time of device implant and had tetraplegia due to a C5 motor/C6 sensory ASIA B spinal cord injury sustained 10 years prior to providing informed consent. This study was conducted under an Investigational Device Exemption from the U.S. Food and Drug Administration and is registered at ClinicalTrials.gov (NCT01894802). The study was approved by the Institutional Review Boards at the University of Pittsburgh and the Space and Naval Warfare Systems Center Pacific. Informed consent was obtained before any study procedures were conducted.

Implantation and electrode arrays

We implanted two sets of microelectrode arrays (Blackrock Microsystems, Inc., Salt Lake City, Utah, Fig. 1B) into the motor and somatosensory cortices. Two microelectrode arrays with 88 wired electrodes (10x10 array, 1.5 mm long, platinum electrodes) were implanted in the hand and arm region of motor cortex in order to decode movement intent. Two additional microelectrode arrays with 32 wired electrodes were implanted in area 1 of somatosensory cortex (6x10 array, 1.5 mm long, sputtered iridium oxide film electrodes) in order to evoke sensations in the fingers of the right hand with intracortical microstimulation (ICMS). Array placement was guided by preoperative imaging, as described previously (*6*, *19*). Area 1 of somatosensory cortex is known to receive cutaneous afferent input from the thalamus and area 3b (*21*) and both of these areas process primarily cutaneous sensory input from peripheral mechanoreceptors in the skin. ICMS in area 1 generates conscious percepts that are primarily tactile in nature including pressure, touch, warm and electrical (*19*). These ICMS-evoked percepts are detectable at latencies within the range of intact tactile percepts (*34*).

Experiment design

The study sessions described here took place between 717 and 738 days after the arrays were implanted. Sessions were conducted twice a week for approximately four hours each day. We tested the two feedback conditions in a block-design over the course of these sessions. For the first four sessions, we used the bidirectional brain-computer interface (BCI) where ICMS-evoked tactile feedback was driven by sensors on the MPL. For the next four consecutive sessions, the same testing protocol was followed, but ICMS was disabled. On each experiment day we trained a new decoder using observation and computer assistance without ICMS (*6*) and then the participant completed three blocks of the sequence task, five blocks of the object transfer task, and one ARAT session using brain control with no computer assistance (Fig. S3).

The participant had unobstructed visual feedback of the computer monitor and robotic hand for all training and testing paradigms, respectively. The participant was free to position his wheelchair to maximize visual feedback from the robotic hand.

Intracortical microstimulation

Stimulation pulse trains consisted of cathodal phase first, current-controlled, charge-balanced pulses delivered at a rate of 100 pulses per second. The cathodal phase was 200 μs long, the anodal phase was 400 μs long, and the amplitude of the anodal phase was set to half the amplitude of the cathodal phase. The phases were separated by a 100-μs interphase period.

To provide somatosensory feedback during real-time prosthesis control, we used electrodes that evoked percepts that were focal to individual fingers, had low detection thresholds, and on

which we could evoke a range of intensities. Detailed descriptions of sensory percepts evoked via ICMS of somatosensory cortex have previously been reported (*19*). For these experiments, we selected electrodes that elicited easily detectable percepts with a clear projected location. One electrode, with a projected field in the proximal interphalangeal joint of the index finger, was mapped to the output of the torque sensor located at the index finger metacarpal phalangeal joint of the Modular Prosthetic Limb (MPL) (*35*) (Fig. 1C and Fig. S1A). Four electrodes with projected fields in either the middle, ring or little finger were mapped to the torque sensor output from the middle finger of the MPL (Fig. 1C and Fig. S1B). Together, the projected fields from the selected electrodes spanned the index, middle, ring and little fingers. MPL finger motor torques were linearly mapped to ICMS current amplitudes (Fig. 1D, Fig. S1, Fig S2, Equation 1) such that increasing the grasp force increased the ICMS current amplitude and, thus, the perceived intensity of the stimulus (*19*). The location of the evoked sensations for given stimulation electrodes are also stable over periods of months to years (*36*). No training was provided for the subject to learn to interpret the sensations evoked by ICMS. All of these factors suggest that while incomplete compared to natural touch, ICMS is able to access neural pathways that were used prior to injury.

Stimulation encoding

For tasks with ICMS, torque sensors located in the motors controlling the Modular Prosthetic Limb (MPL) fingers provided the signal that was used to modulate ICMS pulse train amplitude according to the follow equation:

$$\boldsymbol{A_t} = \left(\frac{\boldsymbol{T_t - T_{min}}}{\boldsymbol{T_{max} - T_{min}}}\right) * (\boldsymbol{A_{max} - A_{min}}) + \boldsymbol{A_{min}} \qquad \text{(Equation 1)}$$

where A_t refers to the commanded pulse train current amplitude at time step t, A_{min} and A_{max} refer to the electrode-specific range of stimulus amplitudes, and T represents the torque sensor data that was being used to relay grasp force. We also set values for the minimum and maximum torque readings, T_{min} and $T_{max,}$ respectively, that corresponded to the minimum and maximum stimulation amplitudes. The selected torque thresholds were 0.1 Nm and 0.5 Nm which corresponded approximately to light touch and strong grasp, respectively. These values were linearly mapped to stimulus amplitudes that ranged from 14 to 64 µA in increments of 4 or 6 µA (Fig. 1D). New torque values were sampled every 20 ms and used to update the pulse train amplitude. This entire process, from sensor data acquisition through to generating new stimulation pulses always occurs within 20 ms, setting an upper bound on the delay between peripheral mechanical events and stimulation in the cortex.

The just-noticeable stimulus amplitude difference on individual electrodes is approximately 15 µA (*19*). Using the chosen stimulus range of 14 to 64 µA, we expect four distinguishable stimulation amplitudes and an ability to reliably distinguish a torque increase of 0.125 Nm.

Neural recording

Extracellular voltage recordings from each electrode in motor cortex were band-pass filtered between 0.3 Hz and 7.5 kHz and digitized at 30,000 samples per second using a NeuroPort signal processor (Blackrock Microsystems, Inc., Salt Lake City, Utah). Electrical artifacts induced by microstimulation were rejected using a combination of digital signal blanking and filtering. During each stimulus pulse the recorded signals were blanked using a sample-and-hold circuit. The signals were then high-pass filtered using a 750 Hz first-order Butterworth filter that minimized the effect of additional transient discontinuities in the signal, enabling fast settling of the wideband signal to

baseline. A spike threshold was set at -4.5 times the root-mean-square of this high-pass filtered signal. Any transient threshold crossings that occurred in the sample immediately after the blanking period were rejected in software. Using this approach, we were able to record single unit activity within 740 μs of the end of a stimulus pulse (*37*).

Decoder training

Threshold crossing signals from extracellular recordings in the motor cortex were used to train a five degree-of-freedom (DoF) decoder that simultaneously and continuously controlled the endpoint velocity of the hand in three-dimensional space, wrist pronation and supination, and hand aperture, with the thumb always opposite the fingers. Since the robotic arm was controlled in an endpoint-velocity mode, individual joint angles in the arm were controlled through an inverse kinematic model with additional constraints to keep the elbow in a natural posture. All 5 DoFs were controlled simultaneously. A 5 DoF control scheme was chosen as it provided a balance between fast training times and a sufficient degree of dexterity to grasp the different objects used in these experiments.

To train the decoder, the participant observed a virtual version of the MPL moving in a 3D environment, as has been described previously (*5*). In this task, the participant was asked to observe and imagine performing the motions of the MPL as the hand was first translated, then oriented, and finally commanded to grasp targets that were randomly presented throughout the workspace using a combination of virtual objects and auditory cues. After observing the completion of 27 trials, which took approximately 7 minutes, an optimal linear estimator decoder was derived using an encoding model that relates neural firing rates to arm kinematics. The encoding model was:

$$f = b_0 + b_x v_x + b_y v_y + b_z v_z + b_\theta v_\theta + b_g v_g \qquad \text{(Equation 2)}$$

where f is the square-root transformed firing rate of a recorded unit, v is a kinematic velocity, and b is a regression coefficient for a given velocity dimension. The dimensions shown in Equation 2 are x, y, and z translation, wrist rotation (θ), and grasp (g). The b coefficients were calculated using linear regression (*38*). Decoder weights were then calculated using indirect optimal linear estimation (*39*).

Next, the participant used the decoder trained from observation data to repeat the training task, however the computer constrained the decoded movement velocities to those that were on the ideal path (*40*). Once this task was completed, a new decoder was trained using the data from the second training set. During task performance, all firing rates were scaled, prior to being decoded, by dividing them by the ratio between the population firing rate during the most recent 300 ms and the population firing rate during decoder calibration. This method of scaling firing rates prior to decoding was developed to compensate for a correlated increase in firing rate across the recorded population that we observe when the prosthetic hand approaches objects (*28*). This scaling allowed the participant to better stabilize the hand near objects in order to grasp them. Ultimately, this velocity decoder was then used, without computer assistance–the decoders and prosthetic arm control systems were naïve to the task goal–to complete the tasks used to evaluate performance.

A new decoder was trained every test day and all decoder training was performed without ICMS. The calibration process took approximately 15 minutes each day.

Decoder testing

Decoder performance was evaluated daily using the physical MPL in a sequence task, where the goal was to acquire instructed combinations of hand endpoint position, wrist orientation and grasp posture (*5*, *6*). A total of 3 sets of 10 trials were performed with the robotic limb without computer assistance to establish the baseline decoder performance accuracy in the absence of objects and, importantly, in the absence of ICMS. A trial was considered successful if the participant was able to place the robotic hand within a position target that was 5 cm in diameter, orient the wrist to within ± 0.25 radians and control the grasp aperture to be at least 80% of the way to maximum flexion or extension of the digits being used.

Functional BCI tasks

We used two different paradigms to quantify the difference between using a motor-only BCI and a bidirectional BCI on the participant's ability to complete functionally relevant tasks. These two tasks, a modified version of the Action Research Arm Test (ARAT) (*11*) (Fig. 1F) and an object transfer task (Fig. 1G), had been successfully performed by the participant with vision as the only source of feedback. Here we directly compared performance with and without ICMS-evoked tactile percepts while vision was always present. All tasks were completed using the MPL. No training or practice time was provided for the subject to learn to control the BCI with ICMS-based sensory feedback.

Action Research Arm Test details and scoring metrics

We conducted a modified version of the ARAT (*11*, *22*), which consisted of nine total tasks: moving eight different objects from the left side of a table to a raised platform located on the right side (Fig. 1F) and one task that simulated pouring water from one glass to another. The objects used in this modified ARAT included four cubes (2.5 cm, 5 cm, 7.5 cm and 10 cm along each edge), a 7.5 cm diameter ball, a rock, and two cylinders (2.5 cm and 1 cm in diameter and 16 cm tall). Additional objects from the ARAT task were too small to be grasped by the MPL. The target platform was 34 x 20.5 cm and was elevated 12 cm off the table surface. The objects started approximately 70 cm away from the target platform. A ninth object from the original ARAT task was also included in which a cup filled with small pieces of paper and plastic–a proxy for water–was placed at the right side of the workspace, and an empty cup was placed 20 cm to the left. The participant's task was to empty the "water" from the cup on the right into the empty cup on the left and replace the originally grasped cup back on the table in an upright position. This task was considered a success if any pieces of paper and plastic landed in the target cup and if the original cup was placed upright on the table.

In all cases, the participant was instructed to complete the task as quickly as possible. The participant had a maximum of two minutes per attempt and had three attempts per object. Each attempt at transferring an object was considered a trial. Trials were timed in real time by experimenters from movement onset to the object being successfully placed on the target platform. Using the ARAT scoring scheme (*11*, *22*), trials were scored on a 3-point system in which a score of zero was awarded if the object was never touched, a score of one was awarded if the object was touched but the participant was unable to complete the task, a score of two was awarded if the task was completed in less than two minutes but more than five seconds, and a score of three was awarded if the task was completed in under five seconds. A score of three is considered to be comparable to able-bodied performance (*11*, *22*). The best score from the three attempts for each object was added together to create a single score for the test. Therefore, with nine objects, a perfect score was 27.

The ARAT score fails to take into account other aspects of performance, such as the total number of completed attempts per object and the actual completion time. Therefore, we recorded video of all trials and measured the time spent reaching for, grasping, and transporting the object. All task phase calculations were done offline, marking individual video frames that spanned each event (Movie S1). We separated the trials into three sequential phases based on robot and object interaction: (1) reach, consisting of movement onset to first object contact; (2) grasp, consisting of first object contact to successful object liftoff; and (3) transport, consisting of object liftoff to object release. The amount of time required to complete each phase was compared across feedback paradigms. For this analysis we included successful trials and did not include trials in which the object was erroneously dropped prior to the platform and re-grasped to continue the trial. There were five such trials in each feedback condition. Water pouring trials were not included as the transport phase is not defined. This resulted in n = 72 trials with ICMS feedback and n = 71 trials without ICMS feedback.

Object transfer task

For the object transfer task, we asked the participant to reach to and grasp a cylindrical object (16 cm tall and 4.3 cm in diameter) with a weighted base placed on the left side of the table, lift the object off of the table, carry it to the target area on the right, and release the object (Fig. 1G). Once the object was placed on the right side of the table, an experimenter returned the object to the start position and the participant repeated the process as many times as possible in two minutes (Movie S4). Performance on this task was measured as the number of times the object was successfully moved across the table in two minutes. Five two-minute trials were completed each day, and this task was always completed immediately following the sequence task.

In the object transfer task, two boundaries were marked on the table that defined a 22.5 cm region where the object was not allowed to touch the table (red area in Fig. 1G). If the object touched the table between these boundaries, the task could be continued by moving the object back to the left side of the table and continuing. For offline analysis, data from the object transfer task was divided into three locations within the workspace, which we called the object, transit, and target zones. We defined these phases using the physical location of the MPL hand. The transit zone consisted of a region 22.5 cm wide and centered on the starting location of the hand at the beginning of a trial (red area in Fig. 1G). The object zone (gray area in Fig. 1G) was located to the left side of the transport zone, while the target zone (green zone in Fig. 1G) was located to the right of the transit zone. These zones were used to compare the amount of time spent in each zone across feedback conditions.

Statistical analysis

Statistical analyses were performed in MATLAB (The MathWorks). Data that were not normally distributed, as determined using Lilliefors test (alpha = 0.05), are reported as medians and interquartile ranges (IQR) and the Wilcoxon rank-sum test was used to assess significance for differences in the median unless otherwise stated. Normally distributed data, as determined using Lilliefors test (alpha = 0.05), are reported as mean ± standard deviation and a two-tailed Student's t-test was used to assess significance for differences in the mean. Specific statistical tests are noted in the text.

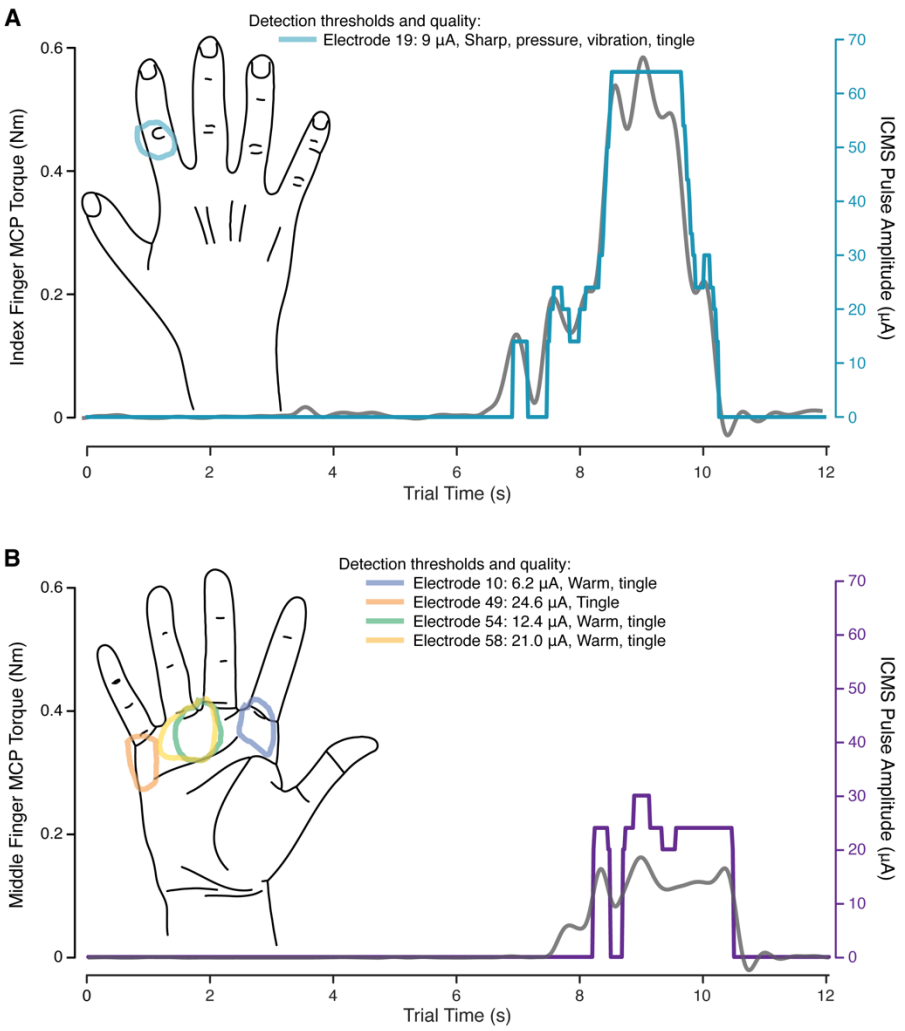


Fig. S1. Example of torque-to-stimulus conversion and electrode properties. An example of the torque and associated stimulation amplitude profiles generated during an ARAT trial with the 7.5 cm cube. This specific trial had the median completion time. (A) The torque sensor data from the index finger motor controlled the stimulation amplitude on an electrode that evoked sensations on the index finger proximal interphalangeal joint. The projected field drawn by the participant is shown on the hand diagram on the left. Detection thresholds and percept quality are listed for the electrode. (B) The torque sensor data from the middle finger motor controlled the stimulation amplitude on four electrodes that evoked sensations at the bases for the 2nd through 5th digits with projected fields shown on the hand diagram. Detection thresholds and percept qualities are listed for each of the electrode. All electrodes used the same stimulation amplitude.

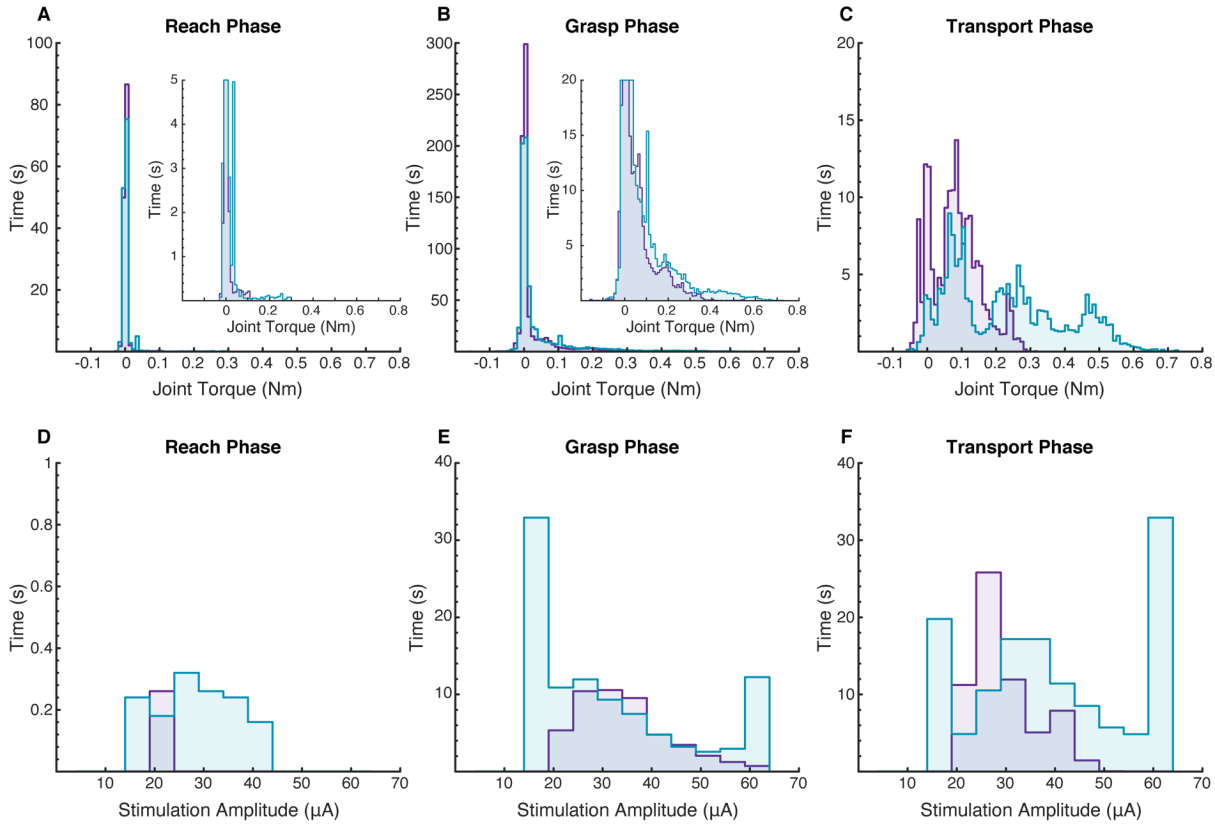


Fig. S2. Histograms of torques and stimulation amplitudes from all successful ARAT trials. Torques and stimulation amplitudes were graded during task performance. Given the large variety of objects and their different mechanical properties, different torques were exerted upon the objects, different stimulation amplitudes therefore were used, and as a result, different percept intensities were evoked in the participant's hand. The y-axes in each plot are scaled differently. The large y-axis values during grasp reflect that most of the trial time occurred in the grasp phase. Index finger torques and associated stimulation amplitudes are shown in blue, while middle finger torques and associated stimulation amplitudes are shown in purple, as in Fig. S1. (A) Torques during the reach phase (trial start to object contact) were very small and were associated with the torque required to move the digit itself. In two trials, small transient torques that exceed the stimulation threshold (0.1 Nm) occurred, but these lasted for < 2 seconds across all the trials. (B) Torques during the grasping phase (object contact to object liftoff) were often small as it still took time for the participant to successfully grasp the object. However, significant torques greater than 0.1 Nm occurred, representing object grasp. Torques exerted by the index finger were generally larger than the torques exerted by the middle finger. (C) During the transport phase (object lift off to object release) torques were much larger, representing the fact that the participant had to hold the object in his grasp until it was released. (D,E,F) Stimulation amplitudes for the reach, grasp and transport phases. Stimulation amplitudes were a linear transformation of the applied torques (Equation 1).

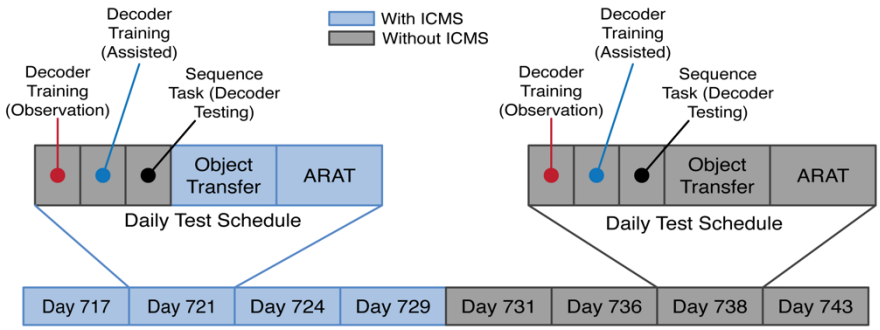


Fig. S3. Experimental timeline showing the experimental schedule within and across days highlighting when ICMS was used (blue) and when it was not (gray). At the beginning of each day a new decoder was trained using a two-step process. An initial decoder was trained using an observation paradigm. Next, the participant controlled the robotic limb with assistance and a new decoder was trained. ICMS was never used during decoding. After the decoder was trained, the participant performed a sequence task to generate baseline control data from which we could determine the overall decoder performance. Finally, the participant performed object transfer trials and then ARAT trials. On the first four experimental days (days 717-729), these functional assessments were conducted with ICMS. On the last four experimental days (days 731-743), functional assessments were conducted without ICMS. Days refer to the time since the electrodes were implanted.

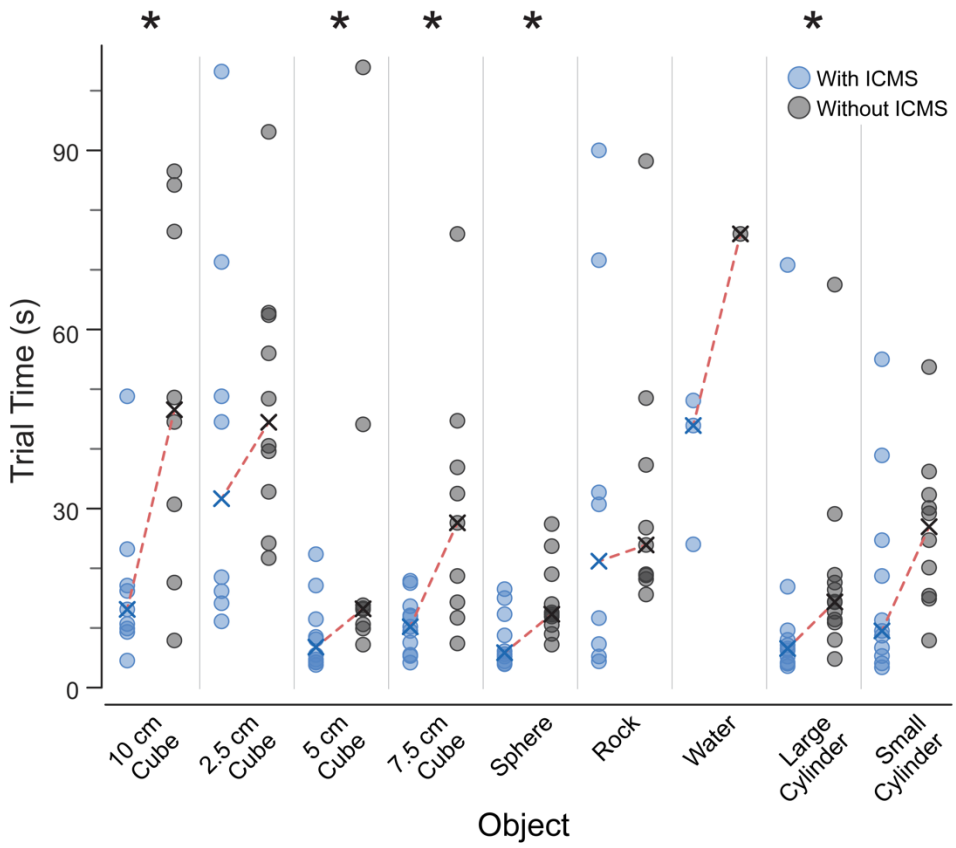


Fig. S4. Effect of ICMS feedback on completion times for individual objects in the ARAT task. Gray dots indicate successful trial times without ICMS feedback while blue dots are successful trial times with ICMS. Median trial times are marked for each object/feedback paradigm with an 'x'. Median times for each object are connected with a red line for visualization. Trial times were significantly faster for five of the nine objects when ICMS feedback was provided (*$p < 0.05$, Wilcoxon rank-sum test).

	Without ICMS			With ICMS				
Object	Median (s)	IQR (s)	N	Median (s)	IQR (s)	N	Wilcoxon U	p-value
10 cm cube	46.6	24.2 – 80.3	8	13.1	9.8 – 18.6	9	13	0.027
2.5 cm cube	44.5	32.8 – 62.0	10	31.6	15.2 – 60.1	8	30	0.408
5 cm cube	13.2	10.3 – 29.0	8	6.8	4.3 – 11.5	10	17	0.043
7.5 cm cube	27.6	13.7 – 38.9	9	10.2	6.0 – 13.2	11	15	0.010
Sphere	12.3	10.9 – 17.8	11	5.9	4.4 – 12.3	10	22.5	0.024
Rock	24.0	18.7 – 40.1	9	21.2	6.3 – 52.2	8	29	0.541
Large Cylinder	14.4	11.2 – 18.3	12	6.6	4.5 – 9.2	11	27.5	0.019
Small Cylinder	27.0	15.4 – 32.3	10	9.5	5.7 – 23.2	11	29.5	0.078
Water		76			24.0, 43.9, 48.1			n/a

Table S1. Successful ARAT trial times for each object. Median and inter-quartile range (IQR) times as well as the number of successful completions (N) are show for each object with and without ICMS-induced tactile sensations. The times for all successful water pouring attempts are listed as there were not enough successfully completed trials to calculate the median and inter-quartile range. The Wilcoxon rank-sum test was used to test for significance.

	Without ICMS		With ICMS				
Phase	Median (s)	IQR (s)	Median (s)	IQR (s)	Change with ICMS (%)	Wilcoxon U	p-value
Reach	2.0	1.5 – 3.2	1.5	1.2 – 2.3	-25.0	1685	0.0004
Grasp	13.3	6.9 – 30.9	4.6	1.8 – 11.8	-65.8	1202	< 0.0001
Transport	2.8	2.0 – 4.2	2.1	1.8 – 2.9	-25.0	1736	0.0009

Table S2. Reach, grasp, and transport times for the ARAT objects. Median and inter-quartile range (IQR) times are shown for each task phase with and without ICMS-induced sensations. Significance was assessed with the Wilcoxon rank-sum test.

Workspace zone	Time without ICMS (s)	Time with ICMS (s)	Change with ICMS (%)	t-statistic	p-value
Object	3.3 ± 1.2	2.3 ± 0.4	-30.3	t38 = 3.3	0.002
Transit	2.1 ± 0.6	2.3 ± 0.3	9.5	t38 = 1.3	0.206
Target	2.8 ± 1.0	2.3 ± 0.5	-17.9	t38 = 2.0	0.048

Table S3. Time spent in the object, transit, and target zones for the object transfer task. Mean and standard deviations of the time taken in each of the three movement phases are shown with and without ICMS-induced sensations Significance was assessed with an unpaired, two-sample Students *t*-test.

	Without ICMS		With ICMS			
Metric	Median	IQR	Median	IQR	Wilcoxon U	p-value
Sequence Task Success Rate	100 %	100 – 100%	95 %	85 – 100 %	40.5	0.025
Translation Velocity	0.23 m/s	0.16 – 0.30 m/s	0.19 m/s	0.12 – 0.27 m/s	8,864,146	< 0.0001

Table S4. Motor performance on the random target sequence task. Median and inter-quartile ranges (IQR) of the success rates (n = 12 scores per condition, 3 repetitions of the task per day for each of 4 experiment test days) and endpoint velocity measures for the random target sequence task (n = 4,533 time bins on testing days without ICMS, n = 4,961 time bins on testing days with ICMS) were used to assess decoder performance in the absence of ICMS on each test day. The sequence task did not involve any object interaction so no ICMS was delivered. Instead, this task served as a metric of overall decoding performance for testing days that were designated as "without ICMS" or "with ICMS" based on the functional tests that were performed following this task. Median success rates were 100% on all 4 testing days without ICMS, while median success rates were 90% for 2 out of the 4 testing days that subsequently used ICMS. Translation velocities were compared to assess whether or not the decoders were biased to perform better on the days with ICMS. The lower velocities on ICMS days suggest that the decoders were not faster so this does not explain the better performance that was observed. A Wilcoxon rank-sum test was used to assess significance as the data were not normally distributed (Lilliefors test, alpha = 0.05).

Movie S1.

Example ARAT trial with ICMS feedback, annotated to indicate task state transitions and illustrate ICMS delivery.

Movie S2.

Fastest ARAT trials for each object and feedback condition.

Movie S3.

ARAT trials for the median completion time for each object and feedback condition. In cases where there were an even number of completed trials, the faster trial is shown in the video.

Movie S4.

Object transfer example trials with and without ICMS feedback. In the full trial, the task lasts for two minutes. The first minute from a trial with the median number of transfers for each feedback condition is used to illustrate performance.

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